Assignment-8: Function and Class Templates

Function Template

1. Write a function template to swap two variables of any data type.

```
Users > Shagun Garg > templatecpp.cpp > main()
#include <iostream>
using namespace std;

template<typename T>
void swapelements(T &a,T &b){
    T temp=a;
    a=b;
    b=temp;
}
int main(){
    int a;int b;
    double x; double y;
    cout<<"Enter integers= ";
    cin>>a;
    cin>>b;
    swapelements(a,b);
    cout<<"Swapped integers :"<<a <<", "<<b<<endl;
    cout<<"Enter double= ";
    cin>>x;
    cin>>y;
    swapelements(x,y);
    cout<<"Swapped doubles :"<<x <<", "<<y<<endl;
    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "c:\Users\Shagun Garg\"
atecpp }
Enter integers= 5
8
Swapped integers :8, 5
Enter double= 5.6
7.8
Swapped doubles :7.8, 5.6
PS C:\Users\Shagun Garg>
```

2. Write a function template to find the minimum element in an array.

```cpp
#include <iostream>
using namespace std;

template<typename T>
T minelement(T &a,T &b){
    return (a<b)?a:b;
}
int main(){
    int a;int b;
    int n;
    cin>>n;
    for (int i=0;i<n;i++){
    cout<<"Enter 1st value= ";
    cin>>a;
    cout<<"Enter 2nd value= ";
    cin>>b;
    cout<<"Minimum element: "<< minelement(a,b)<<endl;
    cout <<endl;
    }
    return 0;
```

```
PS C:\Users\Shagun Garg> cd "c:\Users\Shagu
atecpp }
3
Enter 1st value= 6
Enter 2nd value= 3
Minimum element: 3

Enter 1st value= 3
Enter 2nd value= 9
Minimum element: 3

Enter 1st value= 6
Enter 2nd value= 5
Minimum element: 5

PS C:\Users\Shagun Garg>
```

3. Write a function template to sort an array using bubble sort.

Users > Shagun Garg > templatecpp.cpp > ...

```cpp
#include <iostream>
using namespace std;
template <class T>
void bubbleSort(T arr[], int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = n - 1; j > i; j--) {
            if (arr[j] < arr[j - 1]) {
                swap(arr[j], arr[j - 1]);
            }
        }
    }
}
int main() {
    int n;
    cout << "Enter number of elements: ";
    cin >> n;
    int arr[n];
    cout << "Enter elements:\n";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }
    bubbleSort(arr, n);

    cout << "Sorted array:\n";
    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }
    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "c:\Users\Sha
atecpp }
Enter number of elements: 5
Enter elements:
3
9
6
7
5
Sorted array:
3 5 6 7 9
PS C:\Users\Shagun Garg>
```

4. Write a function template to perform linear search.

```
Users > Shagun Garg > templatecpp.cpp > linearsearch<T>(T [], int, int)
#include <iostream>
using namespace std;
template <class T>
void linearsearch(T arr[], int n, int key) {
    for (int i = 0; i < n - 1; i++) {
            if (arr[i]==key) {
            cout<<"Element " <<key<< " found at index "<<i;
            }
        }
}
int main() {
    int n;
    cout << "Enter number of elements: ";
    cin >> n;
    int k;
    cout<<"Enter key to be found: ";
    cin>>k;
    int arr[n];
    cout << "Enter elements:\n";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }
    linearsearch(arr, n,k);

}
```

```
PS C:\Users\Shagun Garg> cd "c:\Users
atecpp }
Enter number of elements: 5
Enter key to be found: 3
Enter elements:
1
2
3
4
5
Element 3 found at index 2
PS C:\Users\Shagun Garg>
```

5. Design overloaded versions of a function template process() such that:

i)Accepts a single parameter

ii)Accepts two parameters of the same type

iii) Accepts two parameters of different types

Class Template

1. Develop a class template for stack with push and pop operations.

```
Users > Shagun Garg > oops.cpp > ...
#include <iostream>
using namespace std;

template <class T>
class Stack {
    T arr[100];
    int top;
public:
    Stack() {
        top = -1;
    }
    void push(T value) {
        if (top == 99) {
            cout << "Stack Overflow" << endl;
            return;
        }
        arr[++top] = value;
    }
    void pop() {
        if (top == -1) {
            cout << "Stack Underflow" << endl;
            return;
        }
        cout << "Popped element: " << arr[top--] << endl;
    }
    void display() {
        for (int i = top; i >= 0; i--) {
            cout << arr[i] << " ";
        }
        cout << endl;
    }
};
```

```
int main() {
    Stack<int> s;
    s.push(10);
    s.push(20);
    s.push(30);
    s.display();
    s.pop();
    s.display();
}
```

```
PS C:\Users\Shagun Garg> cd "c:\Use
30 20 10
Popped element: 30
20 10
PS C:\Users\Shagun Garg>
```

2. Develop a class template for queue with enqueue and dequeue operations.

```
Users > Shagun Garg > oops.cpp > ...
#include <iostream>
using namespace std;

template <class T>
class Queue {
    T arr[100];
    int front, rear;
public:
    Queue() {
        front = 0;
        rear = -1;
    }
    void enqueue(T value) {
        if (rear == 99) {
            cout << "Queue Full" << endl;
            return;
        }
        arr[++rear] = value;
    }
    void dequeue() {
        if (front > rear) {
            cout << "Queue Empty" << endl;
            return;
        }
        cout << "Deleted element: " << arr[front++] << endl;
    }
    void display() {
        for (int i = front; i <= rear; i++) {
            cout << arr[i] << " ";
        }
        cout << endl;
    }
};
```

```
int main() {
    Queue<int> q;
    q.enqueue(5);
    q.enqueue(10);
    q.enqueue(15);
    q.display();
    q.dequeue();
    q.display();
}
```

```
PS C:\Users\Shagun Garg> cd "c:\
5 10 15
Deleted element: 5
10 15
PS C:\Users\Shagun Garg> 
```

3. Create a class template to store and display a pair of values.

Users > Shagun Garg > oops.cpp > Pair<T1, T2> > second

```cpp
#include <iostream>
using namespace std;

template <class T1, class T2>
class Pair {
    T1 first;
    T2 second;
public:
    Pair(T1 a, T2 b) {
        first = a;
        second = b;
    }
    void display() {
        cout << "First: " << first << endl;
        cout << "Second: " << second << endl;
    }
};
int main() {
    Pair<int, string> p(10, "Hello");
    p.display();
}
```

```
PS C:\Users\Shagun Garg> cd "c:\Users\
First: 10
Second: Hello
PS C:\Users\Shagun Garg> 
```

4. Create a class template for basic arithmetic operations.

Doubly_LL.cpp ● priorityQueue.cpp ● oops.cpp ● oop

Users > Shagun Garg > oops.cpp > Arithmetic<T> > a

```cpp
#include <iostream>
using namespace std;

template <class T>
class Arithmetic {
    T a, b;
public:
    Arithmetic(T x, T y) {
        a = x;
        b = y;
    }
    void add() {
        cout << "Addition: " << a + b << endl;
    }
    void subtract() {
        cout << "Subtraction: " << a - b << endl;
    }
    void multiply() {
        cout << "Multiplication: " << a * b << endl;
    }
    void divide() {
        cout << "Division: " << a / b << endl;
    }
};

int main() {
    Arithmetic<int> obj(20, 5);
    obj.add();
    obj.subtract();
    obj.multiply();
    obj.divide();
}
```

```
PS C:\Users\Shagun Garg> cd "c:\
Addition: 25
Subtraction: 15
Multiplication: 100
Division: 4
PS C:\Users\Shagun Garg> 
```

5. Create a class template to input and display array elements.

```
sers > Shagun Garg > oops.cpp > Array<T> > display()
#include <iostream>
using namespace std;
template <class T>
class Array {
    T arr[100];
    int n;
public:
    void input() {
        cout << "Enter number of elements: ";
        cin >> n;

        for (int i = 0; i < n; i++) {
            cin >> arr[i];
        }
    }
    void display() {
        cout << "Array elements: ";
        for (int i = 0; i < n; i++) {
            cout << arr[i] << " ";
        }
        cout << endl;
    }
};

int main() {
    Array<int> a;
    a.input();
    a.display();
}
```

```
PS C:\Users\Shagun Garg> cd "c:\Users\Shagun Ga
Enter number of elements: 5
10
20
30
40
50
Array elements: 10 20 30 40 50
PS C:\Users\Shagun Garg>
```